In figure 4.5, you can see the OpenTelemetry concept of signal collection, with a strong focus on correlation. Starting from the bottom and moving to the top, the figure shows the flow of the signals from applications and infrastructure, through the OpenTelemetry collector, eventually ingested into backend destinations.

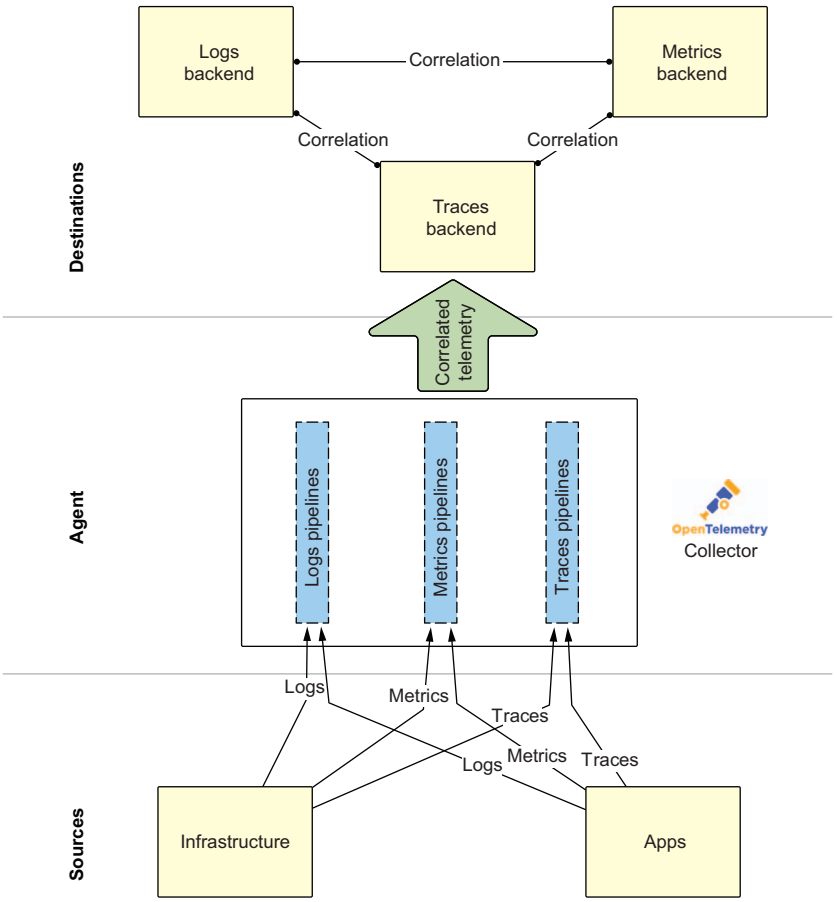


Figure 4.5 Concept of OpenTelemetry collection (from upstream docs; https://opentelemetry.io/docs/reference/specification/logs/overview/)

In the collector, the various signal types (logs, metrics, and traces, currently) are processed in so-called pipelines, dedicated to a signal type. We will discuss this agent's inner workings in greater detail a little later this chapter, but let's first dive into the application code side of the OpenTelemetry house: the instrumentation.